Huawei Access Node QoS Congestion Event Resiliency SOP - KR-ACC-02

Node: KR-ACC-02 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2442

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps to address QoS congestion events (microbursts, interface tail drops, QoS queue starvation) on Huawei Access Node KR-ACC-02 at KR Puram Access site. The failure domain is triggered by critical alarm ALARM_2442. The underlying protocol mechanics responsible for this failure domain are the dynamic bandwidth allocation (DBA) algorithms and the Quality of Service (QoS) policies implemented in the Node's access node controller. The DBA algorithms, such as the Generic Flow Control (GFC) mechanism, dynamically adjust the bandwidth allocation to each user's flow based on their traffic characteristics. However, in scenarios where microbursts or sudden spikes in traffic occur, the DBA algorithms may not be able to adapt quickly enough, leading to QoS queue starvation and interface tail drops. This, in turn, triggers the critical alarm ALARM_2442, indicating a severe congestion event. To prevent such events, this SOP outlines critical steps to diagnose and resolve the root causes of the congestion event, ensuring optimal network performance and reliability.

Pre-requisite Checks

- Verify that the Node KR-ACC-02 is accessible via remote hands at KR Puram Access site.

- Check the status of the redundant link and ensure it is operational.

- Confirm that the Huawei CLI is properly configured for access node operations.

Potential Root Causes

Cause	Probability
Dynamic Bandwidth Allocation (DBA) algorithm malfunction due to unanticipated microbursts or sudden traffic spikes.	High
Inadequate Quality of Service (QoS) policy configuration, leading to QoS queue starvation.	Medium
Interface hardware malfunction or misconfiguration, resulting in interface tail drops.	Low

Diagnostic Commands & Expected Outputs

```
Command: display qam queue, Expected Output: Queue statistics for all classes of service., Description: This command displays the queue statistics for all classes of service, helping to identify potential QoS queue starvation issues.

Command: display interface, Expected Output: Interface statistics, including errors and drops., Description: This command displays the interface statistics, including errors and drops, to help identify potential interface hardware malfunction or misconfiguration issues.
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Command: display dba config, Expected Output: Dynamic Bandwidth Allocation (DBA) configuration., Description: This command displays the DBA configuration, which can help identify potential issues with the DBA algorithm or QoS policies.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy configuration., Command: set qos policy class queue , Description: Verify and adjust the QoS policy configuration to ensure adequate QoS queue management and prevent potential QoS queue starvation issues.

Step 2: Step: Configure Dynamic Bandwidth Allocation (DBA) algorithm., Command: set dba config flow-control-mode , Description: Configure the DBA algorithm to adapt quickly to sudden traffic spikes and prevent potential microburst-induced congestion events.

Step 3: Step: Check and replace interface hardware, if necessary., Command: display interface, Description: Check the interface statistics and error counters to identify potential interface hardware malfunction or misconfiguration issues. If necessary, replace the faulty interface hardware to prevent further congestion events.

Post-Check Verification

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Command: display qam queue, Expected Output: Queue statistics for all classes of service, indicating improved QoS queue management., Description: Verify the queue statistics for all classes of service to ensure improved QoS queue management and prevent potential QoS queue starvation issues.

Command: display interface, Expected Output: Interface statistics, including reduced errors and drops., Description: Verify the interface statistics to ensure reduced errors and drops, indicating successful resolution of potential interface hardware malfunction or misconfiguration issues.

Command: display dba config, Expected Output: Dynamic Bandwidth Allocation (DBA) configuration, indicating improved DBA algorithm performance., Description: Verify the DBA configuration to ensure improved DBA algorithm performance and prevent potential microburst-induced congestion events.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration., Command: restore qos policy class , Description: Restore the original QoS policy configuration to ensure previous QoS queue management settings are preserved.

- Step: Revert DBA algorithm configuration., Command: revert dba config flow-control-mode, Description: Revert the DBA algorithm configuration to its original settings to prevent unintended DBA algorithm performance degradation.

- Step: Replace original interface hardware., Command: install interface , Description: Replace the original interface hardware to ensure previous interface settings and performance characteristics are restored.